

Example #1

Calculate the maximum hydraulic radius (R) and flow depth (y) for a 10 m wide rectangular channel with non-erodible sides with a bed of fine sand (non-colloidal) and a bed slope of 0.0002 m/m. Assume the water is transporting non-colloidal silts/sand and that $n=0.020$.

Solution:

How do we start a problem like this?

Manning's formula:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

Need to find R. We know $S = 0.0002$ and $n=0.020$

What is V?? Were we not given enough information?

There must be a maximum V with will not erode the bed of fine sand...

Idea... design manuals!! Allowable velocity TABLE. (see NEH document on LEARN)

$$V_{all} = 0.46 \text{ m/s}$$

Solve for R

$$R = ((V \cdot n) / S^{0.5})^{3/2} = 0.525 \text{ m}$$

$$R = A/P$$

$$A = W \cdot y$$

$$P = 2y + W$$

Solve...

$$y = (RW) / (W - 2R)$$

$$\text{Depth } y = 0.586 \text{ m}$$

Safety factor??

Table 1. Allowable velocities for different channel materials.

Original material excavated for canals	Mean velocity, for straight canals of small slope, after aging with flow depths less than 3 ft (0.9 m)					
	Clear water, no detritus		Water transporting colloidal silts		Water transporting noncolloidal silts, sands, gravels, or rock fragments	
	ft/s	m/s	ft/s	m/s	ft/s	m/s
Fine sand (noncolloidal)	1.5	0.46	2.5	0.76	1.5	0.46
Sandy loam (noncolloidal)	1.75	0.53	2.5	0.76	2.0	0.61
Silt loam (noncolloidal)	2.0	0.61	3.0	0.91	2.0	0.61
Alluvial silt (noncolloidal)	2.0	0.61	3.5	1.07	2.0	0.61
Ordinary firm loam	2.5	0.76	3.5	1.07	2.25	0.69
Volcanic ash	2.5	0.76	3.5	1.07	2.0	0.61
Stiff clay (very colloidal)	3.75	1.14	5.0	1.52	3.0	0.91
Alluvial silt (colloidal)	3.75	1.14	5.0	1.52	3.0	0.91
Shales and hardpans	6.0	1.83	6.0	1.83	5.0	1.52
Fine gravel	2.5	0.76	5.0	1.52	3.75	1.14
Graded, loam to cobbles (when noncolloidal)	3.75	1.14	5.0	1.52	5.0	1.52
Graded silt to cobbles (when colloidal)	4.0	1.22	5.5	1.68	5.0	1.52
Coarse gravel (noncolloidal)	4.0	1.22	6.0	1.83	6.5	1.98
Cobbles and shingles	5.0	1.52	5.5	1.68	6.5	1.98

Example #2.

An irrigation raceway (open channel) needs to be designed to bring $6.08 \text{ m}^3/\text{s}$ of water from the Rakaia River to dairy farmland 4 km away. The natural slope of the terrain is 0.005 and the channel is to be built through a colloidal alluvial silt soil. The water sourced from the river is transporting colloidal silts. Assume the roughness in the channel bed will be $n = 0.025$ and that you are trying to keep the peak water depth in the channel at or below 0.5 m. Assume the channel will be rectangular (sides lined with brick) and the width will be 8 m.

- i. Calculate a channel design slope.
- ii. Estimate how many drop structures you will need minimize costs.

Solution

flow	6.08
length	4000
Allowable velocity from design table	1.52
slope	0.005
mannings	0.025
Width	8
depth	0.5
Area	4
perimeter	9

Calculate R from A/P

Slope $= V^2 \cdot n^2 / R^{(4/3)}$ (from manning equation)

Slope (allowable) = 0.00426

Actual slope = 0.005

Permissible (allowable) drop = $4000 * 0.00426 = 17.03 \text{ m}$

Actual drop = $4000 * 0.005 = 20 \text{ m}$

Difference 2.97 m

Consider 6 drop structures of 0.5 m height.

(note this problem is setup... in real life you would not be given as much information).

